

1.8 Reaction Kinetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Hard

Time allowed: 10

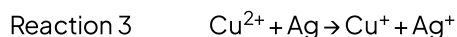
Score: /8

Percentage: /100

Question 1

Photochromic glass is used for some glasses to enable them to work as sunglasses. The glass darkens when exposed to bright light and returns to transparent again when the light is less bright. The darkness of the colour of the glass is related to the concentration of silver atoms.

The following reactions are involved.



Which statement about these reactions is correct?

- A. Ag^+ ions are oxidised in reaction 1.
- B. Cu^+ and Cu^{2+} ions act as catalysts.
- C. Cu^+ ions act as an oxidising agent in reaction 2.
- D. Reaction 2 is the one in which light is absorbed.

[1 mark]

Question 2

When 1 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HCl}$ is added to 10 cm^3 of $0.02 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ pale yellow precipitate forms slowly.

When the experiment is repeated with 1 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HCl}$ and 10 cm^3 of $0.05 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ the precipitate forms more quickly.

Why is this?

- A. When $0.05 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ is used the reactant particles collide more frequently.
- B. When $0.05 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ is used the collisions between reactant particles are more violent.
- C. When $0.05 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ is used the activation energy of the reaction is lower.
- D. When $0.05 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$ is used the reaction proceeds by a different pathway.

[1 mark]

Question 3

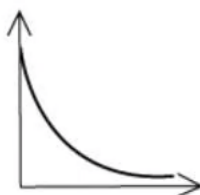
A reaction is zero order with respect to a reactant **A**.

Which concentration–time graph for reactant **A** is the correct shape?

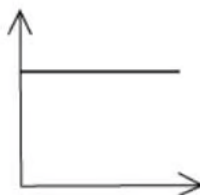
A.



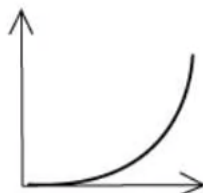
B.



C.



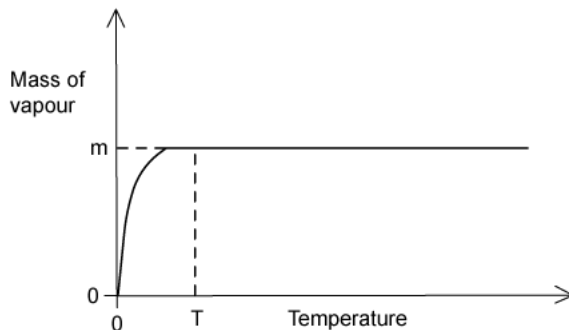
D.



[1 mark]

Question 4

A quantity of solid Y was placed in a previously evacuated vessel, and the apparatus was then held at a series of different temperatures. At each temperature, the mass of Y in the vapour state was calculated from pressure measurements. The results are shown below.



What can be deduced from the diagram?

- 1 The mass of Y used in the experiment was m .
- 2 The pressure of the vapour was constant for all temperatures above temperature T .
- 3 Liquid appeared at temperature T .

A. 1 only

B. 1 and 3

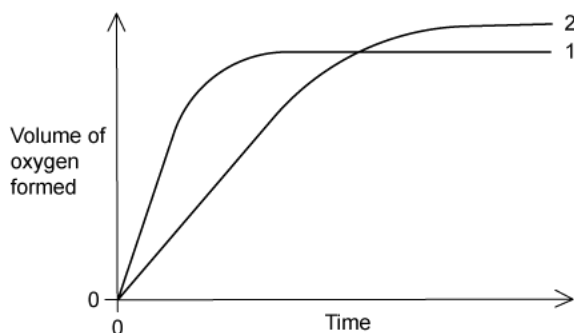
C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 5

On the graph, curve 1 was produced from measuring the decomposition of 100 cm^3 of 1.0 mol dm^{-3} hydrogen peroxide in the presence of a manganese (IV) oxide catalyst.



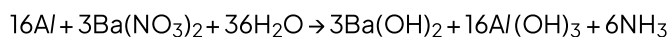
Which alteration to the original experimental conditions would produce curve 2?

- A. Raising the temperature.
- B. Adding more manganese (IV) oxide.
- C. Adding water.
- D. Adding some 0.1 mol dm^{-3} hydrogen peroxide.

[1 mark]

Question 6

The following exothermic reaction occurs during a fireworks display.



The production of sparkler fireworks requires a mixture of barium nitrate powder and aluminium powder to be mixed with water before being glued on to wires. The wire coating must be dry before lighting the fireworks.

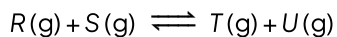
Which conditions would be best to reduce the rate of this reaction during the drying process, and would also keep the aluminium and barium nitrate unchanged?

	pH	Temperature / K
A	7	298
B	14	298
C	7	398
D	14	398

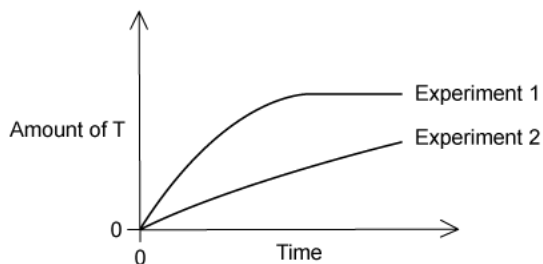
[1 mark]

Question 7

The stoichiometry of a catalysed reaction is shown by the equation below.



Two experiments were carried out in which the amount of product T was measured against time. The results are shown in the diagram below.



Which changes in the conditions from experiment 1 to experiment 2 might explain the results shown?

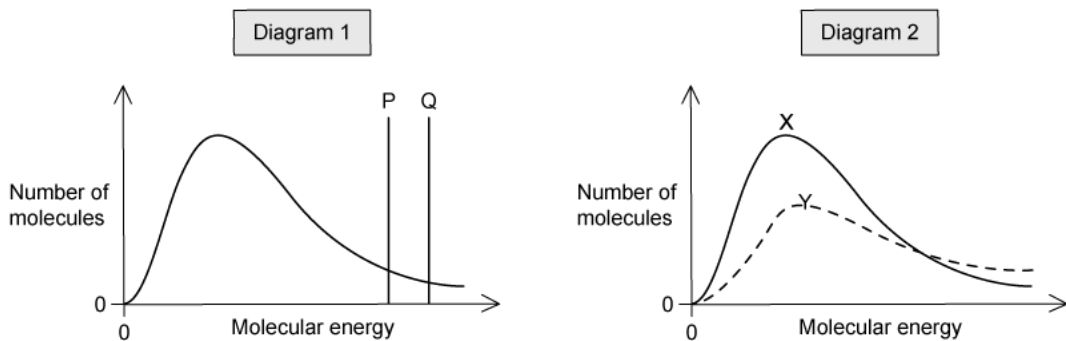
- 1 Product U was continuously removed from the equilibrium mixture
- 2 A different catalyst was used
- 3 Less of reactant R was used

- A. 1 only
 B. 1 and 3
 C. 2 and 3
 D. 1, 2 and 3

[1 mark]

Question 8

Boltzmann distributions are shown in the diagrams.



In diagram 1, one line, P or Q, corresponds to the activation energy in the presence of a catalyst and the other line corresponds to the activation energy of the same reaction in the absence of a catalyst.

In diagram 2, one curve, X or Y, corresponds to a temperature higher than that of the other curve.

Which combination gives the correct curve and line?

	Presence of catalyst	Higher temperature
A	P	X
B	P	Y
C	Q	X
D	Q	Y

[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.
- Copper ions do not form part of the products in the reaction and silver, and chlorine ions are present for the backwards of reaction 1 once the light is removed.

A is incorrect as Ag^+ is reduced, OILRIG, reduction is gain of electrons.

C is incorrect as copper ions are being oxidised and therefore are acting as a reducing agent.

D is incorrect as the reaction that uses the light will need to be reversible in order for the glass to return to original shade once light removed; this will be reaction 1.

2

The correct answer is **A** because:

- Increasing the concentration of substances taking part in a reaction increases the rate of reaction.
- This is because if there are more reacting particles in a mixture, they will collide more frequently.
- The more frequent the collisions, the greater the probability of the reactants have the correct orientation and energy for the reaction to take place.

B is incorrect as the collisions are not more violent when the concentration is changed as there has been no change in energy.

C is incorrect as the concentration does not change the activation energy of the

reaction.

D is incorrect as an alternative pathway would be provided by a catalyst not a change in concentration.

3

The correct answer is **A** because:

- A **zero-order** reaction is one where the rate is constant and independent of the concentration of the reacting substances.
- In this type of reaction, the limiting factor is something other than the concentration of the reactants.
- This means as the reaction progresses the concentration will decrease in a linear relationship.

B is incorrect as this is not a linear relationship and concentration will decrease at a constant rate.

C is incorrect as the concentration will not stay the same throughout the reaction.

D is incorrect as the concentration will not increase.

4

The correct answer is **A** because:

- The question states that the mass of Y is calculated using pressure measurements.
- At point m in the graph, the mass of the vapour does not change.
 - This indicates that all of the solid has vapourised, so statement 1 is correct.

Statement 2 is incorrect as the graph does not show the change in pressure only the mass calculated from the pressure. The vessel may increase in pressure with increasing temperature.

Statement 3 is incorrect as the temperature increases then Y will remain in the gaseous state and not return to a liquid state.

5

The correct answer is **D** because:

- Curve 2 is a slower rate; the gradient of the line is less.
- The initial concentration is less in the mixture as it has been diluted with the addition of the 0.1 mol dm^{-3} hydrogen peroxide.
- The reduction in concentration will reduce the rate of the reaction.
- Over time it will produce a higher volume of oxygen as more H_2O_2 is present to decompose to O_2 .

6

The correct answer is **A** because:

- To reduce the rate of the reaction a lower temperature is used, 298 K.
- To keep the amount of barium nitrate and aluminium constant a lower or neutral pH will be used.
- If the pH is increased more hydroxides will be produced which will result in an increase of aluminium to nitrates ratio.
 - Therefore pH 7 is used.

7

The correct answer is **C** because:

- In experiment 1 as time increases the amount of product T increases until the reactants are used up.
- In experiment 2 the rate of production of T is increased; therefore statement 2 is correct.
- In experiment 2 there is less product T produced; therefore, statement 3 is correct

Statement 1 is incorrect as if one of the products is continuously removed this would increase the forward reaction but still have the same shape for the production of T.

Curve 2 is a slower rate, the gradient of the line is lower.

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7

The correct answer is **C** because:

- In experiment 1 as time increases the amount of product T increases until the reactants are used up.
- In experiment 2 the rate of production of T is increased; therefore statement 2 is correct.
- In experiment 2 there is less product T produced; therefore, statement 3 is correct

Statement 1 is incorrect as if one of the products is continuously removed this would increase the forward reaction but still have the same shape for the production of T.

8

The correct answer is **B** because:

- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.
- In diagram 1 the letter showing the lower activation energy is P.
- At higher temperatures, the Boltzmann distribution moves down and to the right, showing more particles with a higher energy.
- In diagram 2 the line showing this is line Y.

